

IMA 201: Modelling and Simulation

Course Code: IMA 201

Course Name: Modelling and Simulation

L-T-P-C: 2-0-3-3

Prerequisites: Nil

Distribution: IEC (subject area)

Semester: Autumn

Approval: 1st Senate

Objective

To train UG students in engineering to apply their knowledge in mathematics to modelling and simulation, which is extensively used in engineering, and also, to enable them to use available software tools for modelling and simulation.

Details of the Course

- **Introduction:** Definitions of system, environment, input and output variables; State variables and their transition; Hierarchy of knowledge about a system; System identification – structure and parameter identification; Deterministic and stochastic systems; static and Dynamic systems; Objectives of modelling and simulation (4)
- **Modelling of systems with known structure:** Review of conservation laws and the governing equation for heat, mass and momentum transfer, Deterministic models – (a) distributed parameter models in terms of partial differential equations and their solutions and (b) lumped parameter models in terms of differential and difference equations, state space models, transfer functions, block diagrams and subsystems, stability of transfer functions, modelling for control, applications using software tools; Stochastic models (8)
- **Neural Network Modelling of systems only with input-output data base:** Neurons, architecture of neural network, knowledge representation, learning algorithm, Multilayer feed forward network and its back propagation learning algorithm, applications using software tools (6)
- **Fuzzy logic modelling of systems only with broad expert knowledge:** Fuzzy sets, membership functions, fuzzy linguistic variables, 'IF-THEN' rules, Fuzzy Inference systems (FIS) – (i) Mamdani type and (ii) Sugeno (TSK) type, applications using software tools (5)
- **Neuro-fuzzy modelling of systems with limited data base:** adaptive neuro-fuzzy inference system (ANFIS), hybrid learning algorithm using software tools; Co-active neuro-fuzzy system (CANFIS) (2)
- **Optimization and design of systems:** summary of gradient based techniques; Nontraditional optimization techniques – (i) Genetic Algorithm (GA) – coding, GA operators, elitism, application using software tools; (ii) Simulated Annealing (5)

Total contact hours: 30

List of Practicals

1. Programming using software tools for modelling and simulation
2. Find the response of a lumped variable model expressed in terms of transfer function using software tools for inputs of (i) unit step function, (ii) unit impact function and (iii) unit ramp function
3. Use of software tools for simulation
4. Use of software tools for Neural Network
5. Use of software tools for FIS and ANFIS
6. Develop software for temperature control in a furnace
7. Develop software for a given engineering problem involving heat transfer
8. Develop software for a given engineering problem involving mass transfer

Suggested Books

1. B.P. Zeigler, H. Praehofer and T. G. Kim, *Theory of Modelling and Simulation*, Academic Press, Harcourt India Pvt. Ltd., Second Edition (2000)
2. Szekely J.S. & Ray W.H., *Process optimization with applications metallurgy and chemical engineering*, Wiley-Interscience (1973)
3. K. Ogata, *Modern Control Engineering*, Prentice Hall of India Pvt. Ltd., Third Edition (2001)
4. J.S.R. Jang, C. T. Sun and E. Mizutani, *Neuro-Fuzzy and Soft Computing*, Prentice Hall of India Pvt. Ltd., Third Edition (2002)
5. Rudra Pratab, *Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers*, Oxford University Press, USA (2009)